

1. Determine the rank of the matrix, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{bmatrix}$

Ans:- $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{bmatrix}$ $R_2 \rightarrow R_2 - R_1$
 $R_3 \rightarrow R_3 - R_1$

$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{bmatrix}$ $R_3 \rightarrow R_3 - R_2$ $\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$

Rank of the matrix $A = 3$

2. If 2 is the eigen value of $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, without using characteristic equation, find other two eigen value.

Ans:- Sum of eigen values = $\text{tr}(A)$
 i.e., $2 + \lambda_1 + \lambda_2 = 11 \Rightarrow \lambda_1 + \lambda_2 = 9$ — ①

Product of eigen values = $|A|$

$\lambda_1 \lambda_2 = 18$ — ②

$\lambda_1 = 18/\lambda_2$, Substitute λ_1 in eq ①.

$\frac{18}{\lambda_2} + \lambda_2 = 9 \Rightarrow \lambda_2^2 - 9\lambda_2 + 18 = 0$

$\therefore$ Values of $\lambda = 6, 3$

11 a) Solve the System of equations by Gauss elimination method.

$$\begin{aligned} a) \quad & x + 2y + 3z = 1 \\ & 2x + 3y + 2z = 2 \\ & 3x + 3y + 4z = 1 \end{aligned}$$

b) Find the eigen values & eigen vectors of,

$$\begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

$$\text{Ans:- } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$A:B = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 3 & 2 & 2 \\ 3 & 3 & 4 & 1 \end{array} \right] \quad \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -1 & -4 & 0 \\ 0 & -3 & -5 & -2 \end{array} \right] \quad R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & 7 & -2 \end{array} \right]$$

$$\rho(A:B) = \rho(A) = n = 3.$$

$\therefore$ System contains unique solution.

$$x + 2y + 3z = 1$$

$$-y - 4z = 0$$

$$7z = -2 \Rightarrow z = \underline{\underline{-2/7}}$$

$$-y - 4z = 0$$

$$-y = 4z$$

$$-y = 4 \times \frac{-2}{7} = \underline{\underline{-8/7}}$$

$$\therefore y = 8/7$$

$$x + 2y + 3z = 1 \Rightarrow x + 2 \times \frac{8}{7} + 3 \times \frac{-2}{7}$$

$$x + \frac{16}{7} - \frac{6}{7} = 1 \Rightarrow x = 1 - \frac{10}{7} = \underline{\underline{-3/7}}$$

$$X = \underline{\underline{\begin{bmatrix} -3/7 \\ 8/7 \\ -2/7 \end{bmatrix}}}$$

$$b) A = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

So the characteristic equation is, ~~$\lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$~~

$$\lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$$

$$\lambda^3 - 12\lambda^2 + (15 + 8 + 16)\lambda - 28 = 0$$

$$\lambda^3 - 12\lambda^2 + 39\lambda - 28 = 0$$

$\therefore$ the eigen values are $\lambda = 1, 7, 4$

To find eigen vector corresponding to $\lambda = 1$, Put $\lambda = 1$ in eqn ①.

$$A - \lambda I = \begin{bmatrix} 4-1 & 2 & -2 \\ 2 & 5-1 & 0 \\ -2 & 0 & 3-1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 3 & 2 & -2 \\ 2 & 4 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow R_2 + R_3$$

$$\sim \begin{bmatrix} 3 & 2 & -2 \\ 0 & 4 & 2 \\ -2 & 0 & 2 \end{bmatrix} \quad R_3 \rightarrow 3R_3 + 2R_1$$

$$\sim \begin{bmatrix} 3 & 2 & -2 \\ 0 & 4 & 2 \\ 0 & 4 & 2 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 3 & 2 & -2 \\ 0 & 4 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} \rho(A) &= 2 \quad n=3 \\ \therefore n-r &= \underline{\underline{1}} \end{aligned}$$

$$3x_1 + 2x_2 - 2x_3 = 0$$

$$~~4x_2 + 4x_3~~ \cdot 4x_2 + 2x_3 = 0$$

$$\text{Put } x_2 = k$$

$$4k = -2x_3 \Rightarrow x_3 = \underline{\underline{-2k}}$$

$$3x_1 + 2k + 4k = 0 \Rightarrow 6k = -3x_1 \Rightarrow x_1 = \underline{\underline{-2k}}$$

$$X_1 = \begin{bmatrix} -2k \\ k \\ -2k \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix}_{k=1} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

To find eigen vector corresponding to $\lambda = 7$, Put $\lambda = 7$ in eqn

$$A_2 = \begin{bmatrix} -3 & 2 & -2 \\ 2 & -2 & 0 \\ -2 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow R_2 + R_3$$

$$\sim \begin{bmatrix} -3 & 2 & -2 \\ 0 & -2 & -4 \\ -2 & 0 & -4 \end{bmatrix} \quad R_3 \rightarrow 3R_3 - 2R_1$$

$$\sim \begin{bmatrix} -3 & 2 & -2 \\ 0 & -2 & -4 \\ 0 & -4 & -8 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} -3 & 2 & -2 \\ 0 & -2 & -4 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} \rho(A) &= 2, n=3, \quad n-r = 3-2 = \underline{\underline{1}} \\ &\text{Infinite solutions} \end{aligned}$$

$$-3x_1 + 2x_2 - 2x_3 = 0$$

$$-2x_2 - 4x_3 = 0$$

$$\text{Put } x_2 = k$$

$$-2k = 4x_3 \Rightarrow x_3 = \underline{\underline{-\frac{k}{2}}}$$

$$-3x_1 + 2k - 2 \times \underline{\underline{-\frac{k}{2}}} = 0$$

$$-3x_1 + 2k + k = 0$$

$$-3x_1 = -3k \quad x_1 = \underline{\underline{k}}$$

$$X_2 = \begin{bmatrix} k \\ k \\ -k/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}_{k=2} = \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix}$$

To find eigen Vector corresponding to $\lambda = 4$, Put $\lambda = 4$ in eq ①

$$A_3 = \begin{bmatrix} 0 & 2 & -2 \\ 2 & 1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$\sim \begin{bmatrix} -2 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & 2 & -2 \\ -2 & 0 & -1 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_1 \quad \sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & -2 \\ 0 & 1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 - R_2$$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \quad \rho(A) = 2 \quad n = 3 \quad \therefore n - \rho = 3 - 2 = \underline{\underline{1}}$$

Infinite Solution

$$2x_1 + x_2 = 0$$

$$2x_2 - 2x_3 = 0$$

$$\text{Put } x_2 = k, \quad 2x_1 = -k \Rightarrow x_1 = -k/2$$

$$2k = 2x_3 \Rightarrow x_3 \rightarrow k$$

$$X_3 = \begin{bmatrix} -k/2 \\ k \\ k \end{bmatrix} = \underline{\underline{\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}_{k=2}}}$$

12. a) Find values of λ & μ for which the system of Equations,

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = \mu$$

a) no solution b) unique solution c) infinite solution

(ii) Find the matrix of transformation that diagonalize the matrix.

$$\text{Ans: } A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

$$A:B = \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & \lambda & \mu \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow 2R_2 - 7R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -39 & -56 \\ 0 & 0 & \lambda-5 & \mu-9 \end{array} \right]$$

a) no solution

$$\rho(A:B) \neq \rho(A)$$

$$\lambda = 5 \quad \mu \neq 9$$

b) unique solution

$$\rho(A:B) = \rho(A) = n$$

$$\lambda \neq 5; \mu = \text{any values}$$

c) Infinite solution

$$\rho(A:B) = \rho(A) < n$$

$$\lambda = 5 \quad \mu = 9$$

$$(10) A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

So, the characteristic Equation is,

$$\lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$$

$$\lambda^3 + (-2 + -14 + 4)\lambda - 16 = 0$$

$$\lambda^3 - 12\lambda - 16 = 0$$

$$\lambda = 4, -2, -2$$

To find eigen vector corresponding to $\lambda = 4$, Put $\lambda = 4$ in eq ①

$$\begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

$$\begin{bmatrix} -3 & -3 & 3 \\ 3 & -9 & 3 \\ 6 & -6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{array}$$

$$\sim \begin{bmatrix} -3 & -3 & 3 \\ 0 & -12 & 6 \\ 0 & -12 & 6 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2 \quad \sim \begin{bmatrix} -3 & -3 & 3 \\ 0 & -12 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-3x_1 - 3x_2 + 3x_3 = 0$$

$$-12x_2 + 6x_3 = 0$$

$\rho(A) = 2$ $n = 3 \therefore n - r = 1$
Infinite Solution.

$$x_2 = k, \quad -12k = -6x_3 \Rightarrow x_3 = 2k$$

$$-3x_1 - 3k + 6k = 0 \Rightarrow -3x_1 = -3k \Rightarrow x_1 = \underline{k}$$

$$X_1 = \begin{bmatrix} k \\ k \\ 2k \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \underline{k=1}$$

To find eigen vector corresponding to $\lambda = -2$, Put $\lambda = -2$ in eq ①

$$A_2 = \begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 6 & -6 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$
$$\sim \begin{bmatrix} 3 & -3 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 1 \quad n = 3 \quad \therefore n - r = \underline{\underline{2}}$$

Put 2 variables as constant.

$$3x_1 - 3x_2 + 3x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$X_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad X_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & -1 & 1 \end{bmatrix}$$

$$X^{-1}AX = D$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = D$$

MODULE - 2 (Dec 2019)

- 3) If $f(x,y) = xe^{-y} + 5y$, find the slope of $f(x,y)$ in the direction at $(4,0)$.

Ans: Slope in the x direction; $f_x = e^{-y}$

$$\text{slope at } (4,0) = e^{-0} = \underline{\underline{1}}$$

4. Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, where $z = e^x \sin y + e^y \cos x$

$$z_{xx} + z_{yy} = 0$$

$$z_x = e^x \sin y + e^y \sin x$$

$$= e^x \sin y - e^y \sin x$$

$$z_{xx} = e^x \sin y - e^y \cos x \longrightarrow (1)$$

$$z_y = e^x \cos y + e^y \cos x$$

$$z_{yy} = -e^x \sin y + e^y \cos x \longrightarrow (2)$$

From eqn (1) and (2)

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

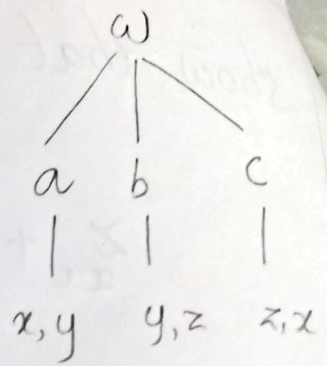
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a) Let f be a differentiable function of three variables and suppose that

$w = f(x-y, y-z, z-x)$, show that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} +$

$$\frac{\partial w}{\partial z} = 0$$

b) Locate all relative extrema of $f(x, y) =$
 $f(x, y) = 4xy - y^4 - x^4$

a) Let $(x-y) = a$, $(y-z) = b$, $(z-x) = c$



$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial w}{\partial c} \cdot \frac{\partial c}{\partial x}$$

$$= \frac{\partial w}{\partial a} \times 1 + \frac{\partial w}{\partial c} \times -1$$

$$= \frac{\partial w}{\partial a} - \frac{\partial w}{\partial c} \longrightarrow \textcircled{1}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial a} \times \frac{\partial a}{\partial y} + \frac{\partial w}{\partial b} \times \frac{\partial b}{\partial y}$$

$$= \frac{\partial w}{\partial a} \times -1 + \frac{\partial w}{\partial b} \times 1$$

$$= -\frac{\partial w}{\partial a} + \frac{\partial w}{\partial b} \longrightarrow \textcircled{2}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial w}{\partial c} \times \frac{\partial c}{\partial z}$$

$$= \frac{\partial w}{\partial b} \times -1 + \frac{\partial w}{\partial c} \times 1$$

$$= -\frac{\partial w}{\partial b} + \frac{\partial w}{\partial c} \longrightarrow \textcircled{3}$$

From eqn $\textcircled{1}$, $\textcircled{2}$ and $\textcircled{3}$

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$b) f(x, y) = 4xy - y^4 - x^4$$

$$f_x = 4y - 4x^3$$

$$f_y = 4x - 4y^3$$

Critical point is obtained by solving

$$f_x = 0 \text{ and } f_y = 0$$

$$f_x = 0$$

$$\Rightarrow 4y - 4x^3 = 0$$

$$y = x^3 \longrightarrow \textcircled{1}$$

$$f_y = 0$$

$$\Rightarrow 4x - 4y^3 = 0$$

$$x = y^3 \longrightarrow \textcircled{2}$$

Substitute eqn $\textcircled{1}$ in $\textcircled{2}$

$$x = (x^3)^3$$

$$x = x^9$$

$$x - x^9 = 0$$

$$x(1 - x^8) = 0$$

$$x = 0$$

$$1 - x^8 = 0$$

$$x^8 = 1 \Rightarrow x = 1 \text{ or } -1$$

4

$$y = x^3$$

$$\text{If } x = 0, y = 0$$

$$x = 1, y = 1$$

$$x = -1, y = -1$$

$\therefore$ The critical points are

$$(0, 0) \quad (1, 1) \quad (-1, -1)$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$f_{xx} = -12x^2$$

$$f_{yy} = -12y^2$$

$$f_{xy} = 4$$

at (0, 0)

$$D = (-12 \times 0) (-12 \times 0) - 4^2$$

$$= -16$$

$$D < 0$$

$\therefore (0, 0)$ is a saddle point.

at (1,1)

$$\begin{aligned} D &= (-12 \times 1^2) \times (-12 \times 1^2) - 4^2 \\ &= (-12 \times -12) - 4^2 \\ &= 144 - 16 \\ &= 128 \end{aligned}$$

$$D > 0$$

$$\therefore f_x = -12$$

$$f_x < 0$$

$\therefore$ (Ans) f has relative maximum at (1,1)

at (-1,-1)

$$\begin{aligned} D &= -12 \times (-1)^2 \times -12 \times (-1)^2 - 4^2 \\ &= 144 - 16 \\ &= 128 \end{aligned}$$

$$D > 0$$

$$f_x = -12, f_x \neq 0$$

$\therefore f$ has relative maximum at (-1,-1)

12. a) Find the local linear approximation L to the fn $f(x, y) = \sqrt{x^2 + y^2}$ at the point $P(3, 4)$. Compare the error in approximating f by L at the point $Q(3.04, 3.98)$ with the distance PQ .

b) The radius and height of a right circular cone are measured with errors of at most 1% and 4% respectively. Use differentials to approximate the maximum percentage error in the calculated volume.

a)
$$f_x = \frac{1}{2\sqrt{x^2 + y^2}} \times 2x$$

$$f_x(3, 4) = \frac{6}{2\sqrt{3^2 + 4^2}}$$

$$= \frac{3}{5}$$

$$f_y = \frac{1}{2\sqrt{x^2 + y^2}} \times 2y$$

$$f_y(3, 4) = \frac{2 \times 4}{2\sqrt{3^2 + 4^2}}$$

$$= \frac{4}{5}$$

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$f(3, 4) = \sqrt{3^2 + 4^2}$$

$$= \sqrt{25} = 5$$

$$L(x, y) = 5 + \frac{3}{5}(3.04 - 3) + \frac{4}{5}(3.98 - 4)$$

$$= 5 + (0.6 \times 0.04) + 0.8 \times -0.02$$

$$= 5 + 0.024 - 0.016$$

$$= 5.008193$$

$$Error = 5.008193 - 5.008$$

distance between the point $(3.04, 3.98)$ and $(3, 4)$

$$= \sqrt{(3.04 - 3)^2 + (3.98 - 4)^2}$$

$$= \underline{\underline{0.04472}}$$

$\therefore$ Error is less than the distance between the points.

b) Volume

$$V = \frac{1}{3} \pi r^2 h$$

Taking logarithm on both sides

$$\log V = \log \left(\frac{1}{3} \pi r^2 h \right)$$

$$= \log \frac{\pi}{3} + \log r^2 + \log h$$

$$= \log \frac{\pi}{3} + 2 \log r + \log h$$

differentiate both side

$$\frac{1}{V} dV = 0 + 2 \times \frac{1}{r} dr + \frac{1}{h} dh$$

$$\frac{dV}{V} \times 100 \leq \left(2 \cdot \frac{dr}{r} + \frac{dh}{h} \right) 100$$

$$\leq (2 \cdot 0.01 + 0.04) 100$$

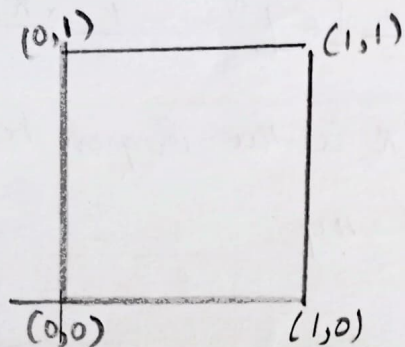
$$\leq 2 + 4$$

$$\leq \underline{\underline{6}}$$

$\therefore$ The maximum % error in calculated volume is 6%

- 5 Find the mass of the square lamina with vertices $(0,0)$ $(1,0)$ $(1,1)$ and $(0,1)$ and density function x^2y

ans



given that $\rho(x,y) = x^2y$

$$M, \text{ mass} = \iint_R \rho(x,y) dA$$

$$= \int_0^1 \int_0^1 x^2y dy dx$$

$$= \int_0^1 \frac{x^2}{2} \left[y^2 \right]_0^1 dx$$

$$= \int_0^1 \frac{x^2}{2} \times 1 dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{6} //$$

- 6) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates.

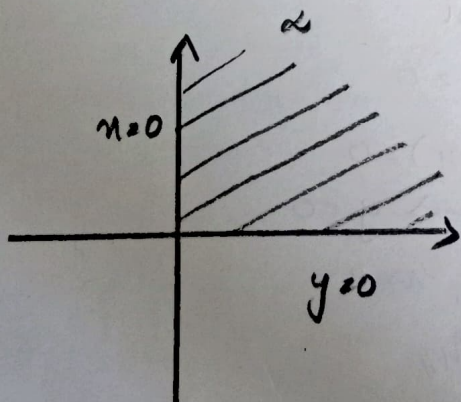
ans: $x^2 + y^2 = r^2$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r: 0 \rightarrow \infty$$

$$\theta: 0 \rightarrow \pi/2$$



$$\int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \int_0^\infty e^{-u} du d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [-e^{-u}]_0^{\infty} d\theta$$

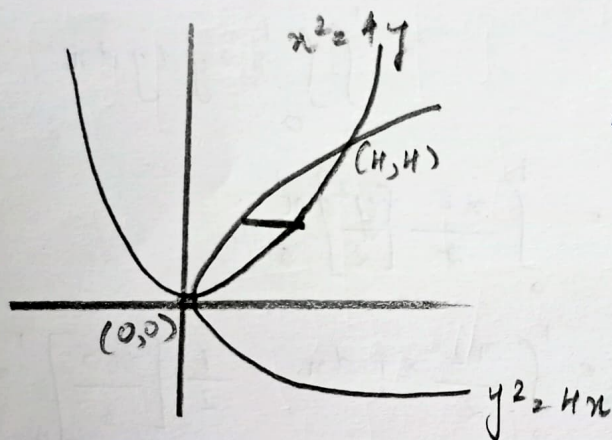
$$\begin{bmatrix} u = r^2 \\ du = 2r dr \\ du/2 = r dr \end{bmatrix}$$

$$= \frac{-1}{2} \int_0^{\pi/2} [e^{-r^2}]_0^{\infty} d\theta = \frac{-1}{2} \int_0^{\pi/2} [e^{-\infty} - e^{-0}] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} 1 d\theta = \frac{1}{2} [\theta]_0^{\pi/2} = \frac{1}{2} \frac{\pi}{2} = \frac{\pi}{4} //$$

15) a) Evaluate $\iint_R y \, dxdy$ where R is the region bounded by the parabola $y^2 = 4x$ and $x^2 = 4y$.

ans:



$$\iint_R y \, dxdy$$

$$\begin{aligned} x^2 &= 4y \\ y &= \frac{x^2}{4} \end{aligned}$$

$$y^2 = 4x$$

$$y = 2\sqrt{x}$$

$$\iint_R y \, dxdy$$

ys.

$$x: \frac{y^2}{4} \rightarrow 2\sqrt{y}$$

$$y: 0 \rightarrow 4$$

$$= \int_0^4 \int_{y^2/4}^{2\sqrt{y}} y \, dxdy$$

$$= \int_0^4 y [x]_{y^2/4}^{2\sqrt{y}} dy$$

$$\Rightarrow \frac{x^2}{4} = 2\sqrt{x}$$

$$x^2 = 8\sqrt{x}$$

$$x^4 = 64x$$

$$x^4 = 64x$$

$$x^4 - 64x = 0$$

$$x(x^3 - 64) = 0$$

$$\text{if } x = 0 \rightarrow y = 0$$

$$\text{if } x^3 = 64, x = 4$$

$$\Rightarrow y = 4 //$$

$$= \int_0^4 y [2\sqrt{y} - y^2/4] dy$$

$$= \int_0^4 y \left[\frac{8\sqrt{y} - y^2}{4} \right] dy = \frac{1}{4} \int_0^4 8y^{3/2} - y^3 dy$$

$$= \frac{1}{4} \left[\frac{8y^{5/2}}{5/2} - \frac{y^4}{4} \right]_0^4$$

$$= \frac{1}{4} \left[\frac{16(2)^5}{5} - \frac{4^4}{4} \right] = \frac{48}{5} //$$

15 b) Use double integral to find the area of the region enclosed b/w the parabola $y = \frac{x^2}{2}$ and the line $y = 2x$. $x: 0 \rightarrow 4$.

g the are of the regions enclosed b/w the parabola

$y = \frac{x^2}{2}$ and the line $y = 2x$

ans: $\int_0^4 \int_{x^2/2}^{2x} dy dx$

$$= \int_0^4 \left[y \right]_{\frac{x^2}{2}}^{2x} dx$$

$$= \int_0^4 \left[2x - \frac{x^2}{2} \right] dx$$

$$= \left[\frac{2x^2}{2} - \frac{x^3}{6} \right]_0^4$$

$$\frac{y = 2x}{\frac{x^2}{2} = 2x}$$

$$x^2 = 4x$$

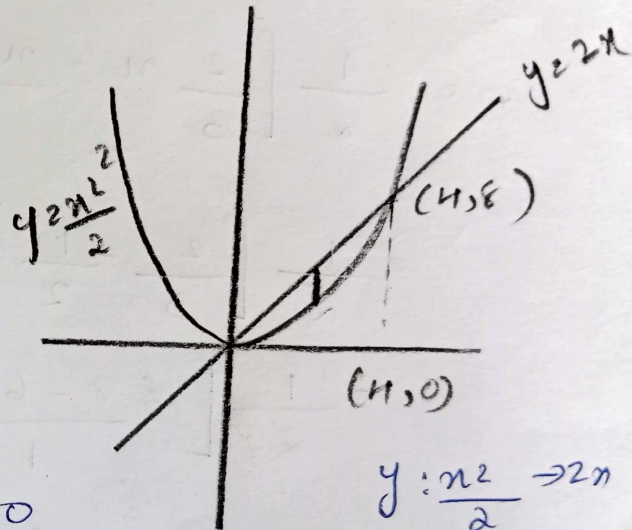
$$x^2 - 4x = 0$$

$$x(x-4) = 0$$

$$x = 0 \quad x - 4 = 0$$

$$x \geq 0, \quad x \leq 4$$

$$y \geq 0, \quad y \leq 8$$



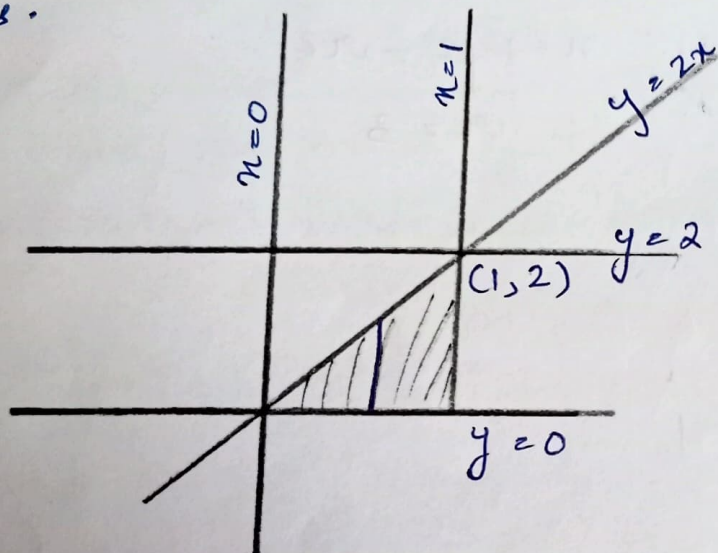
$$y: \frac{x^2}{2} \rightarrow 2x$$

$$x: 0 \rightarrow 4$$

$$= \frac{4^2}{2} - \frac{4^3}{6} = \frac{16}{2} - \frac{32}{3} = \frac{16}{3}$$

16) a. Evaluate $\int_0^2 \int_{y/2}^1 e^{x^2} dx dy$ by reversing the order of integration

ans.



$$x = \frac{y}{2} \rightarrow x = 1$$

$$y: 0 \rightarrow 2 \Rightarrow y = 2$$

x	$y = 2x$
0	0
1	2
2	4

considering vertical strip

$$y: 0 \rightarrow 2x$$

$$x: 0 \rightarrow 1$$

$$\int_0^1 \int_0^{2x} e^{x^2} dy dx = \int_0^1 e^{x^2} [y]_0^{2x} dx$$

$$\left[\begin{array}{l} x^2 = u \\ du = 2x dx \end{array} \right]$$

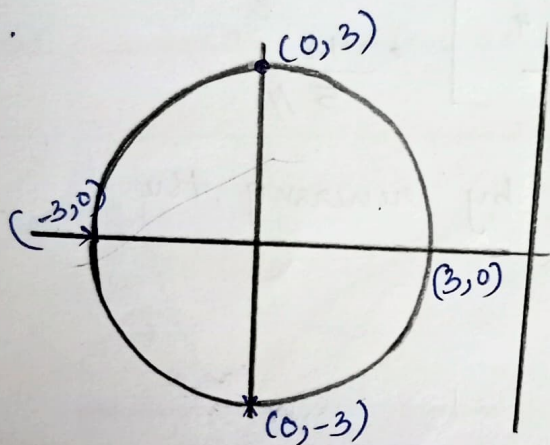
$$= \int_0^1 e^{x^2} [2x] dx$$

$$= \int_0^1 e^u du = [e^u]_0^1 = [e^{x^2}]_0^1$$

$$= [e^1 - e^0] = e - 1 //$$

16b) use triple integrals to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $x=1$ and $x+z=5$

ans.



$$x^2 + y^2 = 9$$

$$z = 1$$

$$x + z = 5$$

put

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\therefore x^2 + y^2 = r^2$$

$$r = 3$$

$$x + z = 5$$

$$z = 5 - x$$

$$= 5 - r \cos \theta$$

$$\text{when } z = 1$$

$$x + 1 = 5$$

$$x = 4$$

$$x : 1 \rightarrow 5 - r \cos \theta$$

$$r : 0 \rightarrow 3$$

$$\theta : 0 \rightarrow 2\pi$$

$$\therefore \int_0^{2\pi} \int_0^3 \int_1^{5-r\cos\theta} r \cdot r \cdot dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 r [z]_1^{5-r\cos\theta} \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 r [5 - r\cos\theta - 1] \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 (4r - r^2 \cos\theta) \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{4r^2}{2} - \frac{r^3}{3} \cos\theta \right]_0^3 d\theta = \int_0^{2\pi} [18 - r \cos\theta] d\theta$$

$$= [18\theta - r \sin\theta]_0^{2\pi} = [18 \times 2\pi - r \sin(2\pi)]$$

$$= 36\pi$$

7. Test the convergence of the series $\sum_{k=1}^{\infty} \frac{k}{2^{k+1}}$

Soln:

$$a_k = \frac{k}{2^{k+1}}$$

$$\lim_{k \rightarrow \infty} \frac{k}{2^{k+1}} = \lim_{k \rightarrow \infty} \frac{1}{2 + 1/k} = \frac{1}{2} \neq 0$$

$\therefore$ series diverges by divergence test.

8. Check the convergence of the series $\sum_{k=1}^{\infty} \frac{1}{k^{k/2}}$

Soln:

$$a_k = \frac{1}{k^{k/2}}$$

$$\rho = \lim_{k \rightarrow \infty} a_k^{1/k} = \lim_{k \rightarrow \infty} \left(\frac{1}{k^{k/2}} \right)^{1/k} = \lim_{k \rightarrow \infty} \frac{1}{k^{1/2}} = 0$$

$\therefore \rho < 1$ $\therefore$ By Root test, $\sum_{k=1}^{\infty} \frac{1}{k^{k/2}}$ converges.

17 a) Find the general term of the series $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \dots$
and use the ratio test to show that series converges.

Soln:

$$a_k = \frac{1 \cdot 2 \cdot \dots \cdot k}{1 \cdot 3 \cdot 5 \cdot \dots (2k-1)}$$

$$a_{k+1} = \frac{1 \cdot 2 \cdot \dots \cdot k(k+1)}{1 \cdot 3 \cdot 5 \cdot \dots (2k-1)(2k+1)}$$

$$\begin{aligned} \rho &= \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{1 \cdot 2 \cdot \dots \cdot k(k+1)}{1 \cdot 3 \cdot \dots (2k-1)(2k+1)} \cdot \frac{1 \cdot 3 \cdot \dots (2k-1)}{1 \cdot 2 \cdot \dots \cdot k} \\ &= \lim_{k \rightarrow \infty} \frac{(k+1)}{(2k+1)} = \lim_{k \rightarrow \infty} \frac{1 + 1/k}{2 + 1/k} = \frac{1}{2} \end{aligned}$$

$\rho < 1 \therefore$ By Ratio test, the given series converges.

17 b) Test whether the following series is absolutely convergent or conditionally convergent. $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$

Soln:

$$u_k = \frac{1}{\sqrt{k(k+1)}} ; u_{k+1} = \frac{1}{\sqrt{(k+1)(k+2)}}$$

$$\lim_{k \rightarrow \infty} \frac{1}{\sqrt{k(k+1)}} = 0$$

$$u_k > u_{k+1}$$

~~By~~ By alternating series test $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$ converges.

Consider the series of absolute values,

$$i.e. \sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}} = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k^2+k}}$$

By informal principle, let $a_k = \frac{1}{\sqrt{k^2+k}}$ and $b_k = \frac{1}{\sqrt{k^2}} = \frac{1}{k}$

$$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k}, \text{ harmonic series } \therefore \text{diverges.}$$

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{\frac{1}{\sqrt{k^2+k}}}{1/k} = \lim_{k \rightarrow \infty} \frac{k}{\sqrt{k^2+k}}$$

$$= \lim_{k \rightarrow \infty} \frac{k}{\sqrt{k^2(1+\frac{1}{k})}} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{k}}} = 1 > 0$$

$\therefore$ By Limit Comparison test, since $\sum_{k=1}^{\infty} b_k$ diverges and finite $\sum_{k=1}^{\infty} a_k$ also diverges i.e. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}}$ diverges.

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k(k+1)}}$ diverges absolutely.

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k(k+1)}}$ is conditionally convergent.

18 a) Test the convergence of $\frac{x}{1 \cdot 2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{3 \cdot 4} + \dots + \frac{x^k}{k(k+1)} + \dots$

Soln:

$$a_k = \frac{x^k}{k(k+1)}$$

$$\rho = \lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k \rightarrow \infty} \frac{|x^{k+1}|}{(k+1)(k+2)} \cdot \frac{k(k+1)}{|x^k|}$$

$$= \lim_{k \rightarrow \infty} \frac{|x| \cdot k}{(k+2)} = |x| \lim_{k \rightarrow \infty} \frac{k}{k+2}$$

$$= |x| \lim_{k \rightarrow \infty} \frac{1}{1+2/k} = |x| \cdot 1 = |x|$$

If $|x| < 1$ the series converges.

If $|x| > 1$, the series diverges.

If $|x| = 1$, test fails.

a) When $x = 1$ or $x = -1$ test fails.

When $x = 1$

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{1}{k^2+k}$$

Let $\sum b_k = \sum_{k=1}^{\infty} \frac{1}{k^2}$, p -series with $p = 2 > 1$
 $\therefore$ converges.

$$\rho = \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{1}{k^2+k} \times k^2 = \lim_{k \rightarrow \infty} \frac{1}{1+1/k} = 1 > 0$$

and finite

$\therefore$ By Limit comparison test, since $\sum \frac{1}{k^2}$ converges
 $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ also converges

When $x = -1$

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \frac{(-1)^k}{k(k+1)}; \quad u_k = \frac{1}{k(k+1)}$$

$$\lim_{k \rightarrow \infty} u_k = \lim_{k \rightarrow \infty} \frac{1}{k(k+1)} = 0$$

$$u_k > u_{k+1}$$

$\therefore$ By Alternating series test $\sum_{k=1}^{\infty} \frac{(-1)^k}{k(k+1)}$ converges.

18 b) Test the convergence of the series $\sum_{k=1}^{\infty} \frac{(k+1)!}{4! k! 4^k}$.

Soln:

$$a_k = \frac{(k+1)!}{4! k! 4^k}$$

$$a_{k+1} = \frac{(k+2)!}{4! (k+1)! 4^{k+1}}$$

$$\begin{aligned} \rho = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{(k+2)!}{4! (k+1)! 4^{k+1}} \cdot \frac{4! k! 4^k}{(k+1)!} \\ &= \lim_{k \rightarrow \infty} \frac{(k+2)}{4 (k+1)} = \lim_{k \rightarrow \infty} \frac{1 + 2/k}{4 + (1/k)} \\ &= \frac{1}{4} < 1 \end{aligned}$$

$\therefore$ Series converges by ratio test.

PART A

Module V (KTU Dec 2019)

9. Find the Taylor series for $f(x) = \cos x$ about $x = \frac{\pi}{2}$ upto third degree term.

Taylor series of $f(x)$ about $x = x_0$ is,

$$f(x) = f(x_0) + \frac{x-x_0}{1!} f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \frac{(x-x_0)^3}{3!} f'''(x_0) + \dots$$

$$f(x) = \cos x, \quad f(\pi/2) = \cos \pi/2 = 0$$

$$f'(x) = -\sin x, \quad f'(\pi/2) = -\sin \pi/2 = -1$$

$$f''(x) = -\cos x, \quad f''(\pi/2) = -\cos \pi/2 = 0$$

$$f'''(x) = \sin x, \quad f'''(\pi/2) = \sin \pi/2 = 1$$

$$f^{(4)}(x) = \cos x, \quad f^{(4)}(\pi/2) = \cos \pi/2 = 0$$

$$\begin{aligned} f(x) &= f(\pi/2) + \frac{x-\pi/2}{1!} f'(\pi/2) + \frac{(x-\pi/2)^2}{2!} f''(\pi/2) + \dots \\ &= -\frac{(x-\pi/2)}{1!} + \frac{(x-\pi/2)^3}{3!} - \frac{(x-\pi/2)^5}{5!} + \dots \end{aligned}$$

10. Find the Fourier half range sine series of $f(x) = e^x$ in $0 < x < 1$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\underline{L=1}$$

$$b_n = \frac{a}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= 2 \int_0^1 e^x \sin(n\pi x) dx$$

$$= 2 \left[\frac{e^x}{1+n^2\pi^2} \left\{ \sin(n\pi x) - n\pi \cos(n\pi x) \right\}_0^1 \right]$$

$$= \frac{2}{1+n^2\pi^2} \left[e^1 (\sin(n\pi) - n\pi \cos(n\pi)) - (e^0 (\sin(0) - n\pi \cos(0))) \right]$$

$$= \frac{2n\pi}{1+n^2\pi^2} [-e(-1)^n + 1]$$

$$= \frac{2n\pi}{1+n^2\pi^2} [1 - e(-1)^n]$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{2n\pi}{1+n^2\pi^2} [1 - e(-1)^n] \sin(n\pi x)$$

Module V (KTU Dec 2019)

19 a) Find the fourier series of periodic function with period 2 which is given below, $f(x) = \begin{cases} -x, & -1 \leq x \leq 0 \\ x, & 0 \leq x \leq 1 \end{cases}$

Hence prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Soln The fourier series expansion of $f(x)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$= \frac{1}{2} \int_{-1}^1 f(x) dx$$

$$= \frac{1}{2} \left[\int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \right]$$

$$= \frac{1}{2} \left[\int_{-1}^0 -x dx + \int_0^1 x dx \right]$$

$$= \frac{1}{2} \left\{ \left[-\frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^1 \right\}$$

$$= \frac{1}{4} [0 - (-1) + 1 - 0]$$

$$= \frac{1}{4} [2] = \underline{\underline{\frac{1}{2}}}$$

$$\left[\begin{array}{l} \text{Here } 2L = +1 - (-1) \\ 2L = 2 \\ L = 1 \end{array} \right]$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^1 f(x) \cos(n\pi x) dx$$

$$= \int_{-1}^0 f(x) \cos(n\pi x) dx + \int_0^1 f(x) \cos(n\pi x) dx$$

$$= \int_{-1}^0 -x \cos(n\pi x) dx + \int_0^1 x \cos(n\pi x) dx$$

$$= -x \times \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{-1}{n\pi} \times \frac{\cos(n\pi x)}{n\pi} \Bigg]_{-1}^0 +$$

$$x \times \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{1}{n\pi} \times \frac{-\cos(n\pi x)}{n\pi} \Bigg]_0^1$$

$$= \frac{-1}{n^2 \pi^2} \cos 0 - \left[\frac{1}{n\pi} \sin(-n\pi) - \frac{1}{n^2 \pi^2} \cos(-n\pi) \right]$$

$$+ \frac{1}{n^2 \pi^2} \cos(n\pi) - \left(\frac{1}{n^2 \pi^2} \cos 0 \right)$$

$$= \frac{-1}{n^2 \pi^2} + \frac{1}{n^2 \pi^2} (-1)^n + \frac{1}{n^2 \pi^2} (-1)^n - \frac{1}{n^2 \pi^2}$$

$$= \frac{-2}{n^2 \pi^2} + \frac{2}{n^2 \pi^2} [-1]^n$$

$$= \frac{2}{n^2 \pi^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^1 f(x) \sin(n\pi x) dx$$

$$= \int_{-1}^0 -x \sin(n\pi x) dx + \int_0^1 x \sin(n\pi x) dx$$

$$\begin{aligned}
&= -x \times \frac{\cos(n\pi x)}{n\pi} - 1 \times \frac{-1}{n\pi} \times \frac{\sin(n\pi x)}{n\pi} \Bigg]_{-1}^0 + \\
&\quad x \times \frac{\cos(n\pi x)}{n\pi} - 1 \times \frac{-1}{n\pi} \times \frac{\sin(n\pi x)}{n\pi} \Bigg]_0^1 \\
&= 0 - \left\{ \frac{-1}{n\pi} \times \cos(-n\pi) \right\} + \frac{-1}{n\pi} \cos(n\pi) - (0) \\
&= \frac{+1}{n\pi} \cos n\pi - \frac{1}{n\pi} \cos n\pi \\
&= 0
\end{aligned}$$

$$\therefore f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi^2} [(-1)^n - 1] \cos n\pi x$$

When $x=0$

$$0 = \frac{1}{2} + \frac{2}{\pi^2} \left[-2 - \frac{2}{3^2} - \frac{2}{5^2} - \frac{2}{7^2} - \dots \right]$$

$$-\frac{1}{2} = \frac{2x-2}{\pi^2} \left[-1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right]$$

$$\frac{-1 \times \pi^2}{2 \times 2 \times -2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

b) Find the half range cosine series for

$$f(x) = \begin{cases} kx, & 0 \leq x \leq L/2 \\ k(L-x), & L/2 \leq x \leq L \end{cases}$$

Soln: Half range Fourier cosine series of $f(x)$ on $(0, L)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where $a_0 = \frac{1}{l} \int_0^l f(x) dx$

$$a_m = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

Given $f(x) = \begin{cases} kx, & 0 \leq x \leq L/2 \\ k(L-x), & L/2 \leq x \leq L \end{cases}$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$= \frac{1}{L} \int_0^L f(x) dx$$

$$= \frac{1}{L} \left[\int_0^{L/2} kx dx + \int_{L/2}^L k(L-x) dx \right]$$

$$= \frac{1}{L} \left\{ \left[k \cdot \frac{x^2}{2} \right]_0^{L/2} + k \left[Lx - \frac{x^2}{2} \right]_{L/2}^L \right\}$$

$$= \frac{1}{L} \left\{ \frac{k}{2} \times \frac{L^2}{4} + k \left[L^2 - \frac{L^2}{2} - \left(\frac{L^2}{2} - \frac{1}{2} \times \frac{L^2}{4} \right) \right] \right\}$$

$$= \frac{1}{L} \left\{ \frac{kL^2}{8} + k \left[\frac{L^2}{2} - \frac{L^2}{2} + \frac{L^2}{8} \right] \right\}$$

$$= \frac{1}{L} \left\{ \frac{kL^2}{8} + k \frac{L^2}{8} \right\}$$

$$= \frac{1}{L} \left\{ 2 \times \frac{kL^2}{8} \right\}$$

$$= \frac{kL}{4}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L} \left\{ \int_0^{L/2} kx \cos\left(\frac{n\pi x}{L}\right) dx + \int_{L/2}^L k(L-x) \cos\left(\frac{n\pi x}{L}\right) dx \right\}$$

$$= \frac{2}{L} \left\{ \left[kx \cdot \frac{\sin\left(\frac{n\pi x}{L}\right)}{\frac{n\pi}{L}} + k \cdot \frac{\cos\left(\frac{n\pi x}{L}\right)}{\left(\frac{n\pi}{L}\right)^2} \right]_0^{L/2} + \right.$$

$$\left. \left[k(L-x) \cdot \frac{\sin\left(\frac{n\pi x}{L}\right)}{\frac{n\pi}{L}} - k \cdot \frac{\cos\left(\frac{n\pi x}{L}\right)}{\left(\frac{n\pi}{L}\right)^2} \right]_{L/2}^L \right\}$$

$$= \frac{2}{L} \left[k \cdot \frac{L}{2} \times \frac{L}{n\pi} \sin\left(\frac{n\pi}{2}\right) + \frac{L^2}{n^2\pi^2} k \cdot \cos\left(\frac{n\pi}{2}\right) - k \cdot \frac{L^2}{n^2\pi^2} (-1)^n + \right.$$

$$\left. - k \cdot \frac{L}{2} \cdot \frac{L}{n\pi} \sin\left(\frac{n\pi}{2}\right) + k \cdot \frac{L^2}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) \right]$$

$$= \frac{2}{L} \left[2k \frac{L^2}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - k \cdot \frac{L^2}{n^2\pi^2} (-1)^n \right]$$

$$= \frac{2Lk}{n^2\pi^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - (-1)^n - 1 \right]$$

$$\therefore f(x) = \frac{kL}{4} + \sum_{n=1}^{\infty} \frac{2Lk}{n^2\pi^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - (-1)^n - 1 \right] \cos\left(\frac{n\pi x}{L}\right)$$

20 a) Find the Fourier series of $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x^2, & 0 < x < \pi \end{cases}$

Soln $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$\boxed{\begin{matrix} 2L = 2\pi \\ L = \pi \end{matrix}}$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^0 f(x) dx + \int_0^{\pi} f(x) dx \right]$$

$$= \frac{1}{2\pi} \left\{ \int_{-\pi}^0 0 dx + \int_0^{\pi} x^2 dx \right\}$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} x^2 dx \right]$$

$$= \frac{1}{2\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$= \frac{1}{2\pi} \times \frac{\pi^3}{3}$$

$$= \frac{\pi^2}{6}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \times \cos(nx) dx + \int_0^{\pi} x^2 \cos(nx) dx \right]$$

$$= \frac{1}{\pi} \left[x^2 \times \frac{\sin(nx)}{n} - 2x \times \frac{1}{n} \times \frac{\cos(nx)}{n} + 2x \times \frac{1}{n^2} \times \frac{\sin(nx)}{n} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[0 + 2\pi \times \frac{1}{n^2} \times \cos(n\pi) + 0 - (0 - 0 + 0) \right] = \underline{\underline{\frac{2}{n^2}(-1)^n}}$$

$$\begin{aligned}
 b_m &= \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} x^2 \sin(nx) dx \right] \\
 &= \frac{1}{\pi} \left[x^2 \times \frac{-\cos(nx)}{n} - 2x \times \frac{-1}{n} \times \frac{\sin(nx)}{n} + 2x \times \frac{-1}{n^2} \times \frac{\cos(nx)}{n} \right]_0^{\pi} \\
 &= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \times \cos(n\pi) - 0 + \frac{2}{n^3} \cos(n\pi) - (0 + 0 + \frac{2}{n^3} \cos 0) \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \times (-1)^n + \frac{2}{n^3} \times (-1)^n - \frac{2}{n^3} \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi^2}{n} (-1)^n + \frac{2}{n^3} [(-1)^n - 1] \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi^2}{n} (-1)^{n+1} + \frac{2}{n^3} [(-1)^n - 1] \right] \\
 &= \frac{\pi (-1)^{n+1}}{n} + \frac{2}{n^3 \pi} [(-1)^n - 1] \\
 \therefore f(x) &= \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2}{n^2} (-1)^n \cos(nx) + \sum_{n=1}^{\infty} \frac{\pi (-1)^{n+1}}{n} + \frac{2}{n^3 \pi} [(-1)^n - 1] \sin(nx)
 \end{aligned}$$

Qo b) Obtain the fourier series expansion for $f(x) = x^2$,
 $-\pi < x < \pi$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

Since $f(x) = x^2$ is an even function, $b_n = 0$.

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi} \times 2 \int_0^{\pi} x^2 dx$$

$$2l = 2\pi$$

$$l = \pi$$

$$\begin{aligned}
 &[\because x^2 \text{ is even}] \\
 &\int_{-\pi}^{\pi} x^2 dx = 2 \int_0^{\pi} x^2 dx
 \end{aligned}$$

$$= \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$= \frac{1}{3\pi} \times \pi^3 = \underline{\underline{\frac{\pi^2}{3}}}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

$$= \frac{2}{\pi} \left[x^2 \times \frac{\sin(nx)}{n} - 2x \times \frac{1}{n} \times \frac{\cos(nx)}{n} + 2 \times \frac{1}{n^2} \times \frac{\sin(nx)}{n} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{2 \times \pi}{n^2} \cos(n\pi) - (0) \right]$$

$$= \underline{\underline{\frac{4}{n^2} (-1)^n}}$$

$$b_n = 0$$

$$\therefore f(x) = \underline{\underline{\frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos(nx)}}$$